

Machine Learning

Project Title: Mercedes-Benz Greener Manufacturing

Project Objective:

Reduce the time a Mercedes-Benz spends on the test bench.

Problem Statement Scenario:

Since the first automobile, the Benz Patent Motor Car in 1886, Mercedes-Benz has stood for important automotive innovations. These include the passenger safety cell with a crumple zone, the airbag, and intelligent assistance systems. Mercedes-Benz applies for nearly 2000 patents per year, making the brand the European leader among premium carmakers. Mercedes-Benz is the leader in the premium car industry. With a huge selection of features and options, customers can choose the customized Mercedes-Benz of their dreams.

To ensure the safety and reliability of every unique car configuration before they hit the road, the company's engineers have developed a robust testing system. As one of the world's biggest manufacturers of premium cars, safety and efficiency are paramount on Mercedes-Benz's production lines. However, optimizing the speed of their testing system for many possible feature combinations is complex and time-consuming without a powerful algorithmic approach.

You are required to reduce the time that cars spend on the test bench. Others will work with a dataset representing different permutations of features in a Mercedes-Benz car to predict the time it takes to pass testing. Optimal algorithms will contribute to faster testing, resulting in lower carbon dioxide emissions without reducing Mercedes-Benz's standards.

Following actions should be performed:

- If for any column(s), the variance is equal to zero, then you need to remove those variable(s).
- Check for null and unique values for test and train sets.
- Apply label encoder.
- Perform dimensionality reduction.
- Predict your test_df values using XGBoost.

Dataset Description:

The dataset is provided as two files, train.csv and test.csv, each containing 4,209 rows. The training file has 378 columns, and the test file has 377 columns. The columns are described below.

- ID is an identifier for each car configuration and is not a predictive feature.
- The variable y is the target. It represents the time, in seconds, that a car spends on the test bench. It appears in the training file only.
- Columns X0 through X385 are 376 anonymized features that encode different permutations of car options and configurations.
- Eight features (X0, X1, X2, X3, X4, X5, X6, and X8) are categorical, and the remaining features are binary.
- Both files are complete and contain no missing values.

Implementation Guidance:

- Remove zero-variance columns, since a feature with zero variance carries no predictive information. The training set contains 12 such columns.
- Check the null and unique values for both sets to confirm the data is complete and to identify the categorical and binary columns.
- Apply the label encoder to the eight categorical columns. Fit the encoder on the combined categories from both the train and test sets, because the test set contains category values that do not appear in the training set. Fitting on the training set alone will cause errors on unseen categories.
- Perform dimensionality reduction on the encoded features using a technique such as PCA, retaining enough components to preserve most of the variance.
- Predict the test set values by training an XGBoost regressor on the reduced training features and the target y , then generating predictions for the test set.